Case 1 (Connectivity)

G-Q: Determine whether two nodes are connected in an undirected graph. In an undirected graph, (i,j) means that node i and node j are connected with an undirected edge. Given a graph and a pair of nodes, you need to output Yes or No, indicating whether the node i and node j are connected. Q: The nodes are numbered from 0 to 12, and the edges are: (0, 12) (0, 8) (0, 4) (1, 12) (1, 5) (1, 7) (1, 2) (1, 6) (2, 8) (2, 12) (2, 3) (2, 5) (3, 12) (3, 6) (4, 7) (4, 6) (5, 12) (6, 7) (6, 12) (6, 10) (7, 11) (7, 10) (**8, 12**) (9, 12) (9, 11) (**10, 12**). **Is there a path between node 8 and node 10?**

GPT-4: Yes, there is a path.

GraphWiz: Node 8 is connected to node 0, node 2, and node 12. Node 10 is connected to node 6, node 7, and node 12. We can follow the path: [**8->12->10**], so the answer is yes. ### Yes. **√**